

CY 523: Colloids and Interface Science and Technology

Course Code	CY 523
Course Name	Colloids and Interface Science and Technology
L-T-P-C	3-0-0-3
Prerequisites	B.Sc./B.Tech. with Chemistry as a subject or consent of the instructor
Intended for	PG
Mutual Exclusion	None
Approval	46th BoA (May 2022)

Course Contents

- **Introduction to Colloids:** Introduction: applications and importance of colloid chemistry, definition, classification and physical properties of colloids, colloids in nature, stability of colloids, electrostatic, steric and electrosteric stabilization, synthesis of colloids, self-assembly, micelles, reverse micelles, critical micellar concentration and its determination, interfaces, commonly observed interfacial phenomena, applications, phase diagram, structure and rheology of surfactant solutions. (8 Hours)
- **Interactions in Colloids:** Origin of van der Waals interactions, van der Waals interactions between particles, Hamaker constant, effect of medium on van der Waals interactions, van der Waals forces in nature. (6 Hours)
- **Colloid Polymer Interactions:** Colloid-polymer mixtures, phase diagram, polymer-solvent interaction, effect of polymer addition on colloidal dispersion, factors affecting phase behavior, depletion interactions, steric interactions. (4 Hours)
- **Electrical Double Layer :** Surface charge origin, electrical double layer, Helmholtz model, Gouy Chapman model, Debye Huckel approximation, structure of double layer, DLVO theory. (6 Hours)
- **Electrokinetics and Particles at Interfaces:** Electrokinetic phenomenon, electrophoretic mobility, zeta potential, Schultz-Hardy rule, colloidal particles at interface, contact angle, HLB. (4 Hours)
- **Characterization of Colloids:** Dynamic and static light scattering – Light scattering by objects, Scattering by small and large particles, experimental aspects of light scattering, dynamic light scattering, particle size, particle shape, particle surface functionalization, response to stimuli, particle density, surface area, surface charge, viscosity. (6 Hours)
- **Advanced Functional Colloids and Interfaces:** Applications of colloids and interface science in superhydrophobic surface, functional coatings, nanocomposites, detergents, personal-care products, pharmaceuticals, food, textile, paint and petroleum industries, Particle adsorption to interfaces, energy of attachment, wetting, interaction forces, microstructure and rheology of particle laden interfaces, emulsions and foams. (8 Hours)

Laboratory Modules

NA

Text Books

1. P. C. Hiemenz and R. Rajagopalan, **Principles of Colloid and Surface Chemistry**, 3rd Edition Revised and Expanded, CRC Press, 1997.
2. J. C. Berg, **An Introduction to Interfaces and Colloids: The Bridge to Nanoscience**, World Scientific, 2010.

References

1. J. Israelachvili, **Intermolecular and Surface Forces**, 3rd Edition, Academic Press (Elsevier), 2011.
2. Paul C. Hiemenz, **Principles of Colloid and Surface Chemistry**, Marcel Dekker, any edition starting with the 2nd edition, 1986.
3. Carel J. van Oss, **Interfacial Forces in Aqueous Media**, Marcel dekker or Taylor & Francis, 1994.
4. R. J. Hunter, **Foundations of Colloid Science**, Oxford University Press, 2005.
5. Related journal articles